

API TEST SUITE: CREATE CUSTOMER API

Endpoint: POST /api/v1/customers | Auth: Bearer Token | Tool: Postman / REST Assured

TC ID	Priority	Test Category	Description
TC_CUST_001	P0	Positive	Verify customer creation with all valid mandatory fields
TC_CUST_002	P0	Validation / Missing Field	Verify API error response when mandatory 'email' field is missing from payload
TC_CUST_003	P1	Validation / Format	Verify API validation error when 'email' format is invalid
TC_CUST_004	P0	Negative / Duplicate	Verify system prevents registration with an already existing email address
TC_CUST_005	P0	Authentication	Verify endpoint rejects requests when Bearer Auth Token is missing

TC_CUST_006	P1	Authentication / Authorization	Verify endpoint response when using an expired or invalid Bearer Token
TC_CUST_007	P1	Boundary	Verify boundary validation when 'firstName' exceeds max allowed length (255 chars)
TC_CUST_008	P2	Positive / Optional Fields	Verify creation with valid optional parameters (phone, address, metadata)
TC_CUST_009	P2	Validation / Payload Format	Verify handling of malformed JSON payload (syntax error)
TC_CUST_010	P1	Negative / Server Error	Verify graceful handling and standard 500 error structure on unexpected database/server failure

Preconditions	Test Steps
1. Active Bearer Token generated 2. Customer email does not exist in DB	1. Set Auth: Bearer <valid_token> 2. Set Header Content-Type: application/json 3. Send POST request with mandatory payload 4. Verify response code and body
1. Active Bearer Token generated	1. Set Auth: Bearer <valid_token> 2. Send POST request omitting 'email' key 3. Verify response status code and field-level error details
1. Active Bearer Token generated	1. Set Auth: Bearer <valid_token> 2. Send POST request with malformed email format ('john.doe@invalid-domain') 3. Inspect error details in response
1. Customer with email 'exists@example.com' exists in DB 2. Active Bearer Token generated	1. Set Auth: Bearer <valid_token> 2. Send POST request using the existing email 3. Confirm conflict response
1. Request constructed without Authorization header	1. Clear Authorization header 2. Send POST request with valid payload 3. Verify unauthenticated status code

1. Token is expired or tampered	1. Set Auth: Bearer <expired_token> 2. Send POST request with valid payload 3. Verify 401/403 status code and exact message
1. Active Bearer Token generated	1. Construct payload with 'firstName' string length of 256 characters 2. Send POST request 3. Verify boundary error handling
1. Active Bearer Token generated 2. Email is unique	1. Set Auth: Bearer <valid_token> 2. Send POST request with all mandatory + optional fields 3. Verify all attributes in response
1. Active Bearer Token generated	1. Send POST request with syntactically malformed JSON (missing closing quote/brace) 2. Verify proper 400 bad request handling without exposing internal stack trace
1. Active Bearer Token generated 2. Mock database service failure triggered on server	1. Configure backend mock/environment to simulate DB connection timeout 2. Send valid POST request 3. Verify system returns structured 500 payload without sensitive stack traces

Test Data (Payload / Request)	Expected Result (Status Code & Response Body)
<pre>{ "firstName": "John", "lastName": "Doe", "email": "john.doe@example.com" }</pre>	Status: 201 Created Response Body: <pre>{ "id": "cust_12345", "firstName": "John", "lastName": "Doe", "email": "john.doe@example.com", "status": "ACTIVE", }</pre>
<pre>{ "firstName": "Jane", "lastName": "Smith" }</pre>	Status: 400 Bad Request Response Body: <pre>{ "errorCode": "INVALID_INPUT", "message": "Validation failed", "errors": [{ "field": "email", }] }</pre>
<pre>{ "firstName": "John", "lastName": "Doe", "email": "john.doe@invalid-domain" }</pre>	Status: 400 Bad Request Response Body: <pre>{ "errorCode": "INVALID_FORMAT", "message": "Validation failed", "errors": [{ "field": "email", }] }</pre>
<pre>{ "firstName": "Alex", "lastName": "Taylor", "email": "exists@example.com" }</pre>	Status: 409 Conflict Response Body: <pre>{ "errorCode": "RESOURCE_ALREADY_EXISTS", "message": "Customer with email 'exists@example.com' already exists." }</pre>
<pre>{ "firstName": "Unauth", "lastName": "User", "email": "unauth.user@example.com" }</pre>	Status: 401 Unauthorized Response Body: <pre>{ "errorCode": "UNAUTHORIZED", "message": "Full authentication is required to access this resource" }</pre>

<pre>{ "firstName": "Expired", "lastName": "Token", "email": "expired.token@example.com" }</pre>	Status: 401 Unauthorized Response Body: <pre>{ "errorCode": "TOKEN_EXPIRED", "message": "JWT token has expired or is invalid" }</pre>
<pre>{ "firstName": "AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA { "firstName": "Sarah", "lastName": "Connor", "email": "s.connor@example.com", "phone": "+14155552671", "address": { "street": "100 Pine St", "city": "San Francisco", "zip": "94111" } } }</pre>	Status: 400 Bad Request Response Body: <pre>{ "errorCode": "FIELD_MAX_LENGTH_EXCEEDED", "message": "Validation failed", "errors": [{ </pre>
<pre>{ "firstName": "Sarah", "lastName": "Connor", "email": "s.connor@example.com", "phone": "+14155552671", "address": { "street": "100 Pine St", "city": "San Francisco", "zip": "94111" } }</pre>	Status: 201 Created Response Body: <pre>{ "id": "cust_67890", "firstName": "Sarah", "lastName": "Connor", "email": "s.connor@example.com", "phone": "+14155552671", }</pre>
<pre>{ "firstName": "BadJSON", "lastName": "Test", "email": "badjson@example.com" // missing closing brace }</pre>	Status: 400 Bad Request Response Body: <pre>{ "errorCode": "MALFORMED_JSON", "message": "JSON parse error: Unexpected character / end-of-input" }</pre>
<pre>{ "firstName": "System", "lastName": "Failure", "email": "sys.fail@example.com" }</pre>	Status: 500 Internal Server Error Response Body: <pre>{ "errorCode": "INTERNAL_SERVER_ERROR", "message": "An unexpected error occurred. Please try again later.", "timestamp": "2026-09-17T20:02:00Z" }</pre>